

Transplant Alchemy 101 - Study Guide

Learning Guide

Objective 1: Understand Who Needs Organs

- Organ and tissue transplantation is required by **patients with end-stage organ failure** to save or significantly improve their lives.
- Thousands of patients are currently on **national waiting lists** for lifesaving organ and tissue transplants.
- Recipients include individuals requiring **hearts** for years of active living, **lungs** to allow unaided breathing, and **kidneys** to save them from dialysis or early death.
- Patients with **liver failure** require transplants to restore life, while those needing a **pancreas** can eliminate insulin dependency.
- Specific indications for kidney transplants include **end-stage renal disease (CKD 5)**.
- Liver transplants are indicated for conditions such as **hepatocellular carcinoma, decompensated cirrhosis, and fulminant hepatic failure**.
- Heart transplants are primarily for those with **end-stage heart failure (NYHA class IV)** and an ejection fraction below 20%.
- Lung transplants are indicated for patients with **advanced lung disease** (such as COPD or Cystic Fibrosis) who have a risk of death exceeding 50% over the next two years.
- Tissue transplants are needed to **restore sight (corneas)**, increase mobility (bone/ligaments), and save lives after **severe burn injuries (skin)**.

Objective 2: Know Where and Why We are Losing Organs

- Organs and tissues are often lost due to a **lack of awareness and ignorance** regarding the donation process.
- Critical shortages are exacerbated by **fear-based understandings**, cultural barriers, myths, and misunderstandings within communities.
- Many potential donations are lost when families automatically say "no" because they have never received **factual information** or were approached at an inappropriate time.
- Organs may be lost because **registered donors** do not share their intentions with their loved ones, as **consent from next of kin** is always required for donation to proceed.
- Misunderstandings regarding **brain death**—such as believing a patient is still "alive" because a ventilator is providing oxygen—can lead families to refuse consent.
- Refusals may stem from concerns about **body disfigurement** during recovery surgery, despite the fact that the body is restored to its original state afterward.

- In the clinical setting, organs can be lost if there is a **failure to identify and refer** a potential donor to a coordinator before death or before treatment is withdrawn.

Objective 3: Know the Team Involved in Transplant

- The **Transplant Coordinator** is a central figure who performs donor assessments, oversees management, and coordinates the entire process.
- A **Tissue Donation Coordinator** is specifically trained to counsel next of kin and obtain consent for tissue-only donations.
- **Two independent registered medical doctors**, who are not part of the transplant team, are legally required to certify brain death.
- The **approaching team** that meets with families often includes a doctor, a nursing sister, a transplant coordinator, and may include a **trauma counselor, translator, or faith representative**.
- The **Organ Donor Foundation (ODF)** serves as the national umbrella body for promoting awareness and managing the donor database.
- The **South African Transplant Coordinators Society (SATCS)** exists to represent, unite, and support all transplant coordinators in the country.
- **Trained tissue recovery technicians** surgically recover tissues in an ethical and respectful manner.
- The **primary treatment team** (physicians and nurses) is responsible for the initial identification and referral of potential donors as part of end-of-life care.

Objective 4: Understand the Basic Legislation Surrounding Organ Donation

- The primary legislation governing these procedures is **Chapter 8 of the National Health Act No. 61 of 2003**.
- There is **no legal requirement for the referral** of a potential donor, but the donation process itself is strictly regulated.
- **Certification of brain death** must be completed by two registered medical doctors, one of whom must have been registered with the **HPCSA for at least five years**.
- Legal death is defined as the time when both doctors have confirmed **brain death** after completing all required clinical tests.
- **Informed consent** must be obtained from the **next of kin** (spouse, parent, child, brother, or sister) or a legal guardian.
- For deaths due to **unnatural causes** (e.g., motor vehicle accidents), additional consent is required from the **state pathologist or district surgeon**.
- Administrative permission to recover organs must be granted by the **medical superintendent or hospital manager** where the recovery occurs.
- By law, the donor's medical aid, estate, and next of kin are **not responsible for any costs** related to the donation after brain death has been declared and consent obtained.

Objective 5: Know When Someone is Eligible for Organ Donation

- Any patient who is declared **brain dead**—irreversible structural brain damage with total loss of brainstem function—is a potential organ donor.
- Common causes of eligibility through brain death include **traumatic head injuries, strokes, or cerebral hypoxia** from drowning or drug overdose.
- There is **no age restriction** for solid organ donation.
- **Donation after Brain Death (DBD)** occurs when a patient on a mechanical ventilator is declared brain dead by two doctors.
- **Donation after Circulatory Death (DCD)** is possible in palliative care settings where treatment is withdrawn and death is imminent.
- **Living organ donors** are eligible if they make a conscious decision to donate a **kidney or a liver segment**.
- **Extended criteria donors** with comorbidities like hypertension, diabetes, or HIV may still be eligible if individual organs are assessed as transplantable.
- **Tissue donation eligibility** is broader; it can occur irrespective of the manner of death and even after the body has been moved to a mortuary.
- **Cornea donors** are generally between **6 and 65 years old**, and poor eyesight or cataracts do not disqualify them.
- **Skin and bone donors** are typically healthy individuals between the ages of **16 and 80** without infectious diseases or cancer.
- **Heart valve donors** are eligible from the age of **6 months up to 55 years**.

References

- **Canter, C. E., Shaddy, R. E., Bernstein, D., Hsu, D. T., Chrisant, M. R. K., Kirklin, J. K., Kanter, K. R., Higgins, R. S. D., Blume, E. D., Rosenthal, D. N., Boucek, M. M., Uzark, K. C., Friedman, A. H., & Young, J. K. (2007).** Indications for heart transplantation in pediatric heart disease: A scientific statement from the American Heart Association Council on Cardiovascular Disease in the Young. *Circulation*, 115(5), 658–676.
- **Critical Care Society of Southern Africa. (2021).** *South African Guidelines on the Determination of Death* (CCSSA Website Version).
- **Mancini, D., & Lietz, K. (2010).** Selection of cardiac transplantation candidates in 2010. *Circulation*, 122(2), 173–183.
- **Mehra, M. R., Canter, C. E., Hannan, M. M., Semigran, M. J., Uber, P. A., Baran, D. A., Danziger-Isakov, L., Kirklin, J. K., Kirk, R., Kushwaha, S. S., Lund, L. H., Potena, L., Ross, H. J., Taylor, D. O., Verschuuren, E. A. M., & Zuckermann, A. (2016).** The 2016 International Society for Heart Lung Transplantation listing criteria for heart transplantation: A 10-year update. *The Journal of Heart and Lung Transplantation*, 35(1), 1–23.
- **Porrett, P. M., Hsu, J., & Shaked, A. (2009).** Late surgical complications following liver transplantation. *Liver Transplantation*, 15(S2), S12–S18.
- **South African Transplant Coordinators Society (SATCS). (n.d.).** *The Organ and Tissue Donation Reference File (Red File)*.
- **Thomson, D., Joubert, I., De Vasconcellos, K., Paruk, F., Mokogong, S., Mathivha, R., McCulloch, M., Morrow, B., Baker, D., Rossouw, B., Mdladla, N., Richards, G. A., Welkovics, N., Levy, B., Coetzee, I., Spruyt, M., Ahmed, N., & Gopalan, D. (2021).** South African guidelines on the determination of death. *South African Medical Journal*, 111(4b), 367–380.
- **Weill, D., Benden, C., Corris, P. A., Dark, J. H., Davis, R. D., Keshavjee, S., Lederer, D. J., Mulligan, M. J., Patterson, G. A., Singer, L. G., Snell, G. I., Verleden, G. M., Zamora, M. R., & Glanville, A. R. (2015).** A consensus document for the selection of lung transplant candidates: 2014—An update from the Pulmonary Transplantation Council of the International Society for Heart and Lung Transplantation. *The Journal of Heart and Lung Transplantation*, 34(1), 1–15.
- **AMBOSS GmbH. Transplantation.** AMBOSS Knowledge Library. <https://next.amboss.com/us/article/gn0Fsg?q=transplantation>. Accessed February 10, 2026